

Community Organisation

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* Basics of communities

1) connected hypothesis

→ A community corresponds to a connected subgraph

2) Density Hypothesis

→ ~~comm~~ communities correspond to locally dense neighbours of a network.

* Internal degree (k^{int}) → set of ~~nodes~~ ^{links} that connect to other nodes of the same community.

→ If $k^{\text{int}} = 0$, then the node must be assigned to a different community.

* External degree (k^{ext}) → set of links that connect to the rest of the network.

→ If $k^{\text{ext}} = 0$, then the community is a good one for the node.

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* Intra-cluster Density

$$s^{int}(c) = \frac{\text{No' of internal links of } c}{N_c (N_c - 1) / 2}$$

Max Possible internal links

Higher value $\rightarrow$ Nodes are highly interconnected.

* Inter-cluster Density

$$s^{ext}(c) = \frac{\text{No' of inter-cluster links of } c}{N_c (N - N_c)}$$

$\rightarrow$ Measures connections from the community to the rest of the network.

Low value $\rightarrow$ strong well defined community

* Key Property of Good communities

$$s^{int}(c) \gg \langle K \rangle \gg s^{ext}(c)$$

$\rightarrow$ Internal connections are much denser than Average

$\rightarrow$ External connections are much weaker than Average

- * clique → set of nodes where every node is connected to every other node.
- fully connected subgraph.

Maximum cliques $s^{\text{int}}(c) = 1$

- Triangles (3 nodes) are common, but larger cliques (5, 6 -- nodes) are rare.
- communities don't necessarily correspond to complete subgraphs as many nodes don't directly link to each other.
- Finding cliques - computationally → NP complete
Extensive Problem

* Strong community :- $k_i^{\text{int}}(c) > k_i^{\text{ext}}(c)$

→ Internal links > External links for Every Node

* Weak community :- $\sum_{i \in c} k_i^{\text{in}}(c) > \sum_{i \in c} k_i^{\text{out}}(c)$

Total Internal links > Total External links

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* Graph Partitioning

→ Dividing a Network into smaller groups (subgraphs) while minimizing connections b/w them.

* Kernighan - Lin Algorithm

Goal → Split a network into 2 equal groups while reducing the connections b/w them.

Step 1 → Divide the graph into two equal groups (A & B)

Step 2 → Evaluate swaps for every pair (one from each group), check how swapping affects cut size (Inter links)

Step 3 → Swap the pair that reduces cut the most, or have the least increase if no improvement.

Step 4 → Repeat until each node is swapped once, keeping the best result.

* Graph bisection

→ Divide N nodes into 2 groups of size N_1 & N_2

No. of ways $\Rightarrow \frac{N!}{n_1! n_2!} \left\{ \text{choosing } n_1 \text{ nodes from } N \right\}$

If $n_1 = n_2 = \frac{N}{2}$ then partitions $\approx \frac{2^N}{\sqrt{N}}$ { exponential }

* Hierarchical clustering

similarity Matrix → Measures how similar two nodes are
→ Built from Adjacency matrix

Group Nodes based on similarity, Two Algorithms:-

1) Agglomerative Algorithm

- Merges nodes & communities with high similarity
- Start with each node as its own cluster & Repeatedly merge the most similar nodes / clusters.

2) Divisive Algorithm

- Start with all nodes in one cluster & Repeatedly remove weak links & split into smaller groups.

* Dendrogram (Tree)

- A tree diagram showing how clusters form
- Bottom → Individual Nodes
- Top → one Big cluster
- cutting Tree horizontally → ~~common~~ gives community partitions

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* Properties of Hierarchical Networks

1) scale free Property

Degree Distribution follows :- Power law $\{ P(K) \sim K^{-\alpha} \}$

Many Nodes $\rightarrow$ small Degree

Few Hubs $\rightarrow$ very High Degree

2) clustering coefficient Independent of N

$C(N) = \text{constant}$

$\rightarrow$ As the Network grows overall clustering remains constant.

3) scaling clustering coefficient

$c(K) \sim K^{-1}$

Small K Nodes $\rightarrow$ High clustering coeff.

$\rightarrow$ Highly interlinked compact communities

High K Nodes $\rightarrow$ small clustering coeff.

(Hubs) $\rightarrow$ connect independent communities.

* Random Hypothesis

→ Randomly wired networks are not expected to have a community structure.

* Modularity

$$M = \sum_{c=1}^{nc} \left[\frac{l_c}{L} - \left(\frac{k_c}{2L} \right)^2 \right]$$

- 1) optimal partition that maximises modularity
- 2) suboptimal partition - positive modularity (not max)
- 3) Negative Modularity - If we assign each node to a different community.
- 4) zero Modularity - Assigning all nodes to the same community

* Modularity is size dependent.

* Maximal Modularity Hypothesis

→ The partition with the maximum modularity offers the best community structure.

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* Modularity based community Identification

* Greedy Technique [Newman]

Step 1 - Assign each node to a community $\{N \text{ nodes} \rightarrow N \text{ communities}\}$

Step 2 - Take all pairs of communities that are connected.
for each pair, compute: ΔM (change in modularity) if merged.

Step 3 - Take the pair with largest +ve ΔM & Merge them into a single community

Step 4 - Repeat until everything becomes one community.

Step 5 - Record for each step & select the partition which is maximal.

~~* overlapping communities~~

greedy Algorithm $\begin{cases} \rightarrow \text{Scalability issue} \\ \rightarrow \text{Not fast, Not successful at maximizing } M. \end{cases}$

Better Algorithms :-

1) Fast Modularity Algorithm

2) Louvain Method

Teacher's Sign

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* clique Percolation

- Two k -cliques (complete graph of k nodes) are considered adjacent if they share $k-1$ nodes.
- A k clique community is the largest connected subgraph obtained by the union of all adjacent k -cliques.
- other k -cliques that can not be reached from a particular clique correspond to other clique communities.